

# TiVCT<sub>x</sub> MXene — Single-/Few-Layer Powder

Research-grade two-dimensional transition-metal carbide powder

TiVCT<sub>x</sub> MXene is supplied as a delaminated single-/few-layer powder derived from TiVAIC. The material has a representative layer count of ≤5. The grade is supplied for electrochemical, energy-storage, conductive-composite, interface and broader two-dimensional-material research.

## KEY FEATURES

- Delaminated single-/few-layer powder derived from TiVAIC
- Nominal lateral size: 200–1000 nm
- Typical conductivity of 1000 ± 500 S/cm
- Surface terminations: –O, –OH, –F, –Cl

## POTENTIAL RESEARCH APPLICATIONS

- Electrochemical and energy-storage material research
- Conductive-composite research
- Interface and two-dimensional-material research

## STORAGE & HANDLING

Store dry in a tightly closed container. Vacuum or inert-atmosphere storage is recommended. Minimize air and moisture exposure.

## QUALITY / DOCUMENTATION

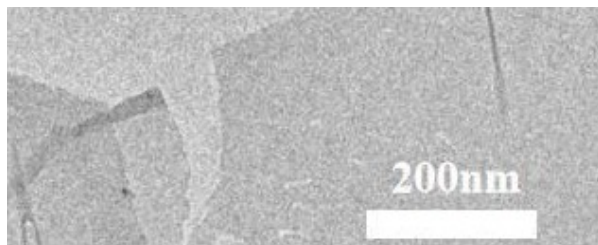
Surface termination distributions vary with synthesis and lot. Quantitative surface composition can be discussed when required.

## TECHNICAL SPECIFICATIONS

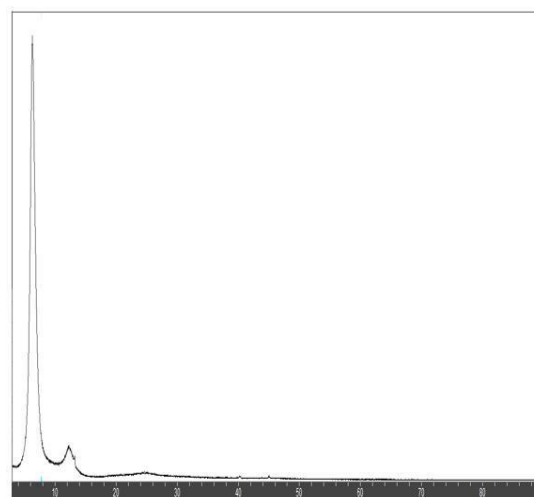
| PROPERTY                | TYPICAL / REPRESENTATIVE VALUE                         |
|-------------------------|--------------------------------------------------------|
| Material                | TiVCT <sub>x</sub> MXene                               |
| Precursor               | TiVAIC                                                 |
| Product form            | Single-/Few-Layer Powder                               |
| Preparation             | LiF/HCl etching followed by intercalation/delamination |
| Layer structure / count | ≤5 layers                                              |
| Nominal lateral size    | 200–1000 nm                                            |
| Typical conductivity    | 1000 ± 500 S/cm                                        |
| Surface terminations    | –O, –OH, –F, –Cl                                       |
| Recommended storage     | Dry; vacuum or inert atmosphere                        |

*Conductivity values are representative technical values. Measurement results can vary with sample preparation, packing, film preparation and measurement geometry.*

## REPRESENTATIVE CHARACTERIZATION



Representative microscopy; not lot-specific.



XRD. Representative characterization; not lot-specific.  
Non-data annotation removed; diffraction trace and axes otherwise unchanged.

WINIGEN MATERIALS

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Representative technical values. Properties can vary with synthesis and lot. Lot-specific documentation and additional analytical details are available subject to review.